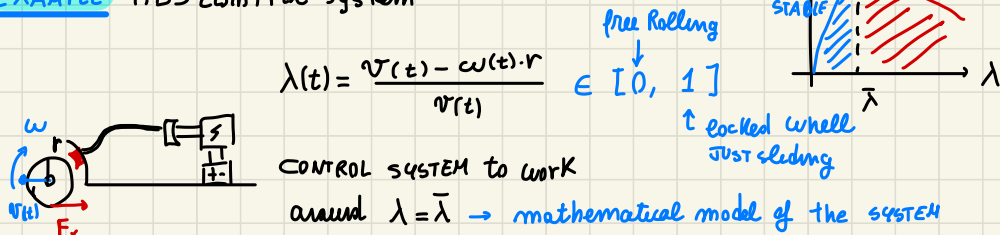


MIDA 2 Recap (learning for control of dynamical systems)

- system model / Identification on Black Box approach
- variable estimation

EXAMPLE ABS control system



SYST IDENTIF. / MODELLING

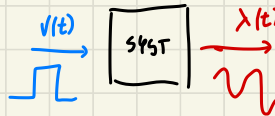
- white Box approach
- Black Box approach

VARIABLE ESTIMATION

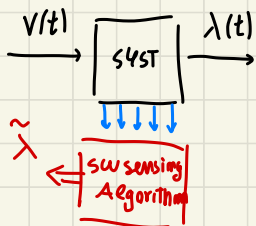
$v(t)$? HOW TO MEASURE

- GPS
- OPTICAL sensors

NOT 100% Available



BUILD MODEL from DATA



- (PRO)
- measure unmeasurable variable
 - cost-saving
 - No maintenance

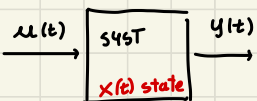
(CONS): Alg. must be tuned

Ch 2.1 B.B model Identification In Time domain, In STATE SPACE

WITH NON PARAMETRICAL APPROACH

(DISCRETE TIME systems) REPRESENTATIONS:

1. STATE SPACE (INTERNAL)



$$\begin{cases} x(t+1) = Fx(t) + Gu(t) \\ y(t) = Hx(t) + Du(t) \end{cases}$$

SISO of ORDER n

$$\begin{cases} F: n \times n & G: n \times 1 \\ H: 1 \times n & D: 1 \times 1 \end{cases}$$

Asymp. stable

IF all Eig(F) are inside the U.C $|Eig(F)| < 1$

! S.S representation is not unique, just a matrix transform $H' = HT^{-1}$
through a matrix T n x n invertible describe same syst $F' = TFT^{-1}$
 $G' = TG$

2. T.F (external) from I/O time domain description, using the Z OPERATOR

$\mathcal{Z}^1(x(t)) = x(t+1) \rightarrow$ LINEAR, $\Rightarrow$ we bring the time domain (t) description into Z domain ordinary

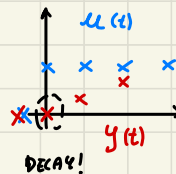
$\mathcal{Z}^{-1}(x(t)) = x(t-1)$ RECURSIVE

$$y(t) = W(z) u(t) = \frac{B(z)}{A(z)} z^{-k}$$

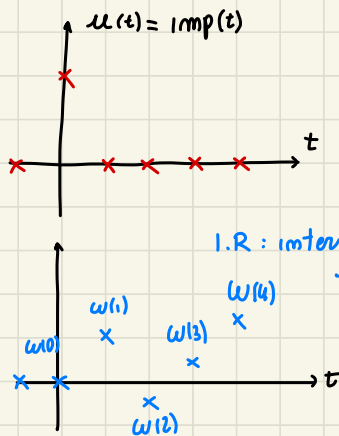
delay both in neg / pos Z powers!

RATIONAL FUNCT of z

if $k \geq 1 \Leftrightarrow$ SYSTEM IS STRICTLY PROPER and represent the response delay to an input step



3. I.R (convolution of Input with I.R)



We can prove easily that given a general input $u(t)$, the output $y(t)$ of the system is

$$y(t) = \sum_{k=0}^{+\infty} w(k) u(t-k)$$

I.R: internal property of the system, invariant

TRANSFORMATION

from S.S to T.F

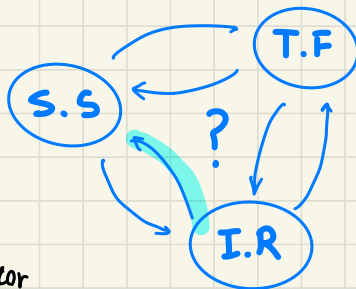
given

$$\begin{cases} x(t+1) = F x(t) + G u(t) \\ y(t) = H x(t) \end{cases} \rightarrow \text{Z operator}$$

$$y(t) = \underbrace{H(zI - F)^{-1}G}_{W(z)} u(t)$$

⚠️ careful with OBS/REACH

you can have an m ORDER system BUT maybe see from T.F JUST $m^* < m$ states! $\Rightarrow$ dynamic "INVISIBLE" externally



► **T.F to S.S** in general to a unique T.F correspond an equivalent S.S description of the system, we consider one special realization, the "CONTROL" one:

$$W(z) = \frac{b_0 z^m + b_1 z^{m-1} + \dots + b_{m-1}}{z^m + a_1 z^{m-1} + \dots + a_m}$$

$$F = \begin{bmatrix} 0 & 1 & 1 & & 0 \\ 0 & 0 & 1 & \ddots & 0 \\ & 0 & & \ddots & 1 \\ -a_m & -a_{m-1} & \dots & -a_1 & 0 \end{bmatrix} \quad G = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$H = [b_{m-1} \ b_{m-2} \ \dots \ b_0] \quad D = [0]$$

► **from T.F to I.R** we have multiple ways: one possible is to use the **LONG DIVISION**

$$W(z) = \frac{B(z)}{A(z)} \rightarrow \begin{array}{l|l} B(z) & A(z) \\ \hline & \leftarrow \text{you highlight the } W(z) \text{ I.R} \\ & \text{obtaining an I.R description, as an ALL-ZEROS T.F} \end{array}$$

$$y(t) = w(0)u(t) + w(1)u(t-1) + w(2)u(t-2) + \dots$$

OR you can Recognize particular series from T.F When Dealing With

DIGITAL FILTERING

$$\left\{ \begin{array}{ll} W(z) = \frac{z^{-1} + \frac{1}{2} z^{-2}}{1 + \frac{1}{4} z^{-1}} & \text{I.I.R. Recursiveness, the I.R never goes to } \infty \\ W(z) = z^{-1} + \frac{1}{4} z^{-2} + \frac{1}{5} z^{-3} & \text{F.I.R. NOT Recursive, I.R goes to } \infty \text{ after } t=3 \text{ (ALL ZEROS T.F)} \end{array} \right.$$

► **from I.R to T.F**

by definition given a DISCRETE TIME signal $s(t)$, $s(t) = 0 \ t < 0$ we define its Z TRANSFORM as and IT can be proven that

$$Z(s(t)) = \sum_{k=0}^{+\infty} z^{-k} (s(k))$$

$$W(z) = Z(w(t)) = \sum_{t=0}^{+\infty} z^{-t} w(t)$$

T.F is the Z TRANSFORM of the I.R

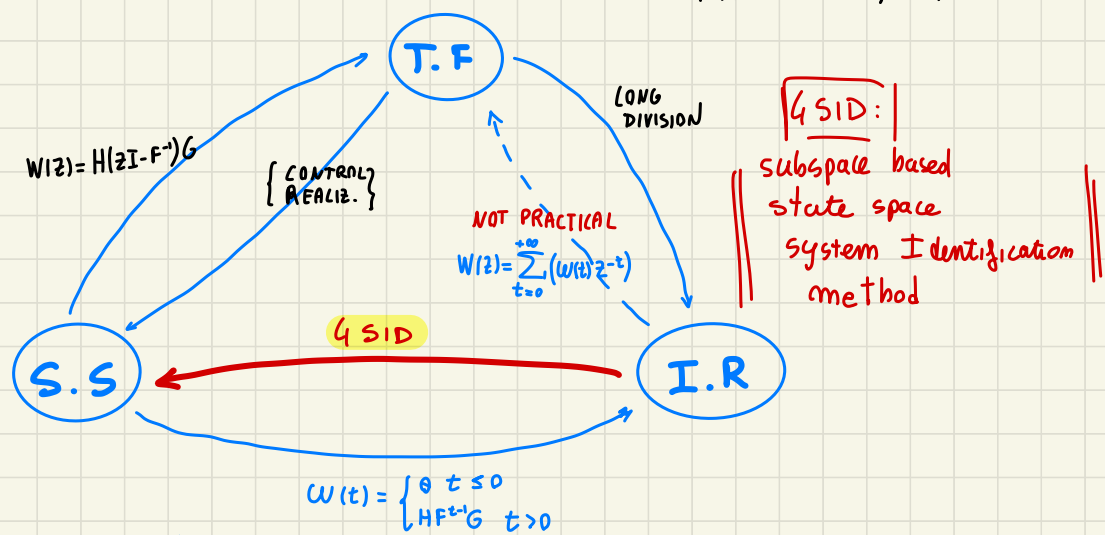
❗ NOT usable in PRACTICE! because you need all the I.R of the system! and IR must be available NOISE FREE

from S.S to I.R

given $\begin{cases} x(t+1) = Fx(t) + Gu(t) \\ y(t) = Hx(t) \end{cases}$ and I.C $\begin{cases} x(0) = 0 \\ y(0) = 0 \end{cases}$

we can find out $\begin{cases} x(1) = Fx(0) + Gu(0) = Gu(0) \rightarrow y(1) = HGu(0) \\ x(2) = FGx(0) + Gu(1) \rightarrow y(2) = HFGx(0) + HGu(1) \\ x(3) = F^2Gx(0) + FGx(1) + Gu(2) \rightarrow y(3) = HF^2Gx(0) + HFGx(1) + HGu(2) \dots \end{cases}$

in general $\begin{cases} w(0) = 0 \\ w(t) = HF^{t-1}G \quad t > 0 \end{cases}$ $y(t) = 0u(t) + HGu(t-1) + HFGu(t-2) + HF^2Gu(t-3) + \dots$



(DEFINITIONS)

OBSERVABILITY the system is fully OBSERVABLE from the output $y(t)$

IF and only IF the OBS MATRIX

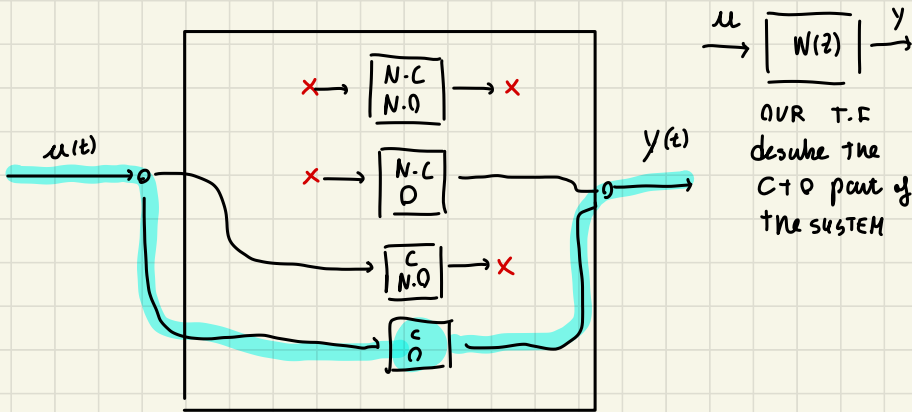
- this means that from $y(t)$ we can fully see the states description, in fact it depends on OUTPUT (H) and STATES (F)

$O = \begin{bmatrix} H \\ HF \\ HF^2 \\ \vdots \\ HF^{m-1} \end{bmatrix}$ is full rank $\text{Rank}(O) = m$

CONTROLLABILITY system is fully CONTROLLABLE from the input $u(t)$ IF and ONLY IF the CONTR/REACH MATRIX $R = [G \quad FG \quad F^2G \quad \dots \quad F^{m-1}G]$ is full rank $\text{Rank}(R) = m$

- this means that by changing the input $u(t)$ you control the full states, in fact it depends on INPUT (G) and STATES (F)

NOTICE from a general system, you can have 4 PARTS

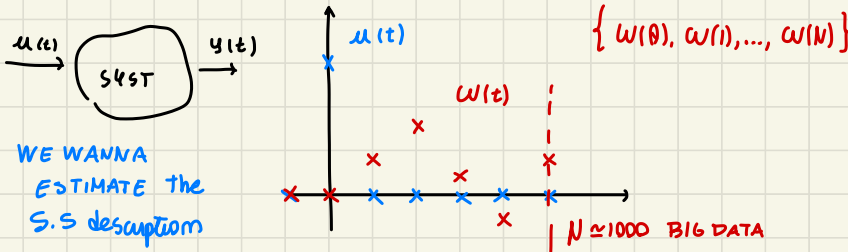


• **HANKEL MATRIX of ORDER m** GSID is based on a special MATRIX built from the I.R dataset $\{w(0), w(1), \dots, w(N)\}$

$$H_m = \begin{bmatrix} w(1) & w(2) & \dots & w(m) \\ w(2) & w(3) & \dots & w(m+1) \\ \vdots & \vdots & \ddots & \vdots \\ w(m) & \dots & \dots & w(2m-1) \end{bmatrix}_{m \times m} = \begin{bmatrix} HG & HFG & \dots & HF^{m-1}G \\ HFG & HF^2G & \dots & HF^mG \\ \vdots & \ddots & \ddots & \vdots \\ HF^{m-1}G & \dots & \dots & HF^mG \end{bmatrix} = Q \cdot R$$

↑ FACTORIZATION

GSID from measurements



WE WANNA
ESTIMATE the
S.S description

Hyp: Noiseless measurements of I.R (IDEAL)

4SID ALGORITHM, NOISLESS ideal case

once collected the DATA of I.R $\{w(1), w(1), w(2), \dots, w(m)\}$

1. SYSTEM ORDER identification

through the HANKEL MATRIX

$$H_1 = [w(1)] \quad \text{rank} = 1 \quad \checkmark$$

$$H_2 = \begin{bmatrix} w(1) & w(2) \\ w(2) & w(3) \end{bmatrix} \quad \text{rank} = 2 \quad \checkmark$$

$\vdots$

$$H_m = \begin{bmatrix} & & & \\ & & & \\ & & & \end{bmatrix} \quad \text{rank} = m \checkmark \quad \leftarrow \text{SYSTEM ORDER is } \textcircled{m}$$

$$H_{m+1} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \quad \text{rank} = m \text{ STOP}$$

2. FACTORIZATION of H_{m+1}

$$O_{m+1} = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^m \end{bmatrix} \begin{matrix} O_1 \\ \\ \\ O_2 \end{matrix} \quad R_{m+1} = \begin{bmatrix} G & FG & \dots & F^m G \end{bmatrix} \begin{matrix} \\ \\ \\ \text{extended obs.} \end{matrix}$$

$\begin{matrix} \xleftarrow{m} & \xleftarrow{m+1} \end{matrix}$

$\begin{bmatrix} O_{m+1} \end{bmatrix} \begin{matrix} \uparrow \\ \downarrow \end{matrix} \begin{bmatrix} R_{m+1} \end{bmatrix}$

$\begin{matrix} m & m+1 \end{matrix}$

$\begin{matrix} \text{extended REACH} \end{matrix}$

3. S.S matrix identification

$$\hat{H} = O_{m+1} \begin{bmatrix} 1 \\ \vdots \end{bmatrix}$$

$$\hat{G} = R_{m+1} \begin{bmatrix} \vdots \\ 1 \end{bmatrix}$$

$$O_2 = O_1 \cdot F \Rightarrow \hat{F} = O_1^{-1} O_2$$

FINISH ALG.

- CONSTRUCTIVE ALG.
NONPARAMETRIC

identified $\hat{F}, \hat{H}, \hat{G}$ from I.R

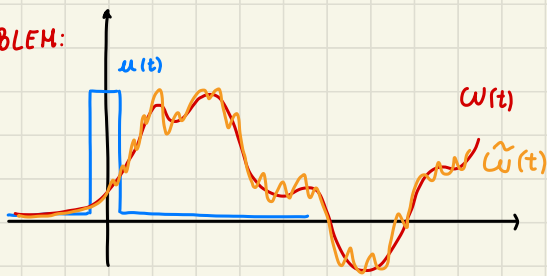
(!) We JUST uses $2m \ll N$ measurements ... just a small noise make the algorithm totally WRONG! USELESS



EXTEND to NOISY MEASUREMENTS

Thanks to Numerical Algebra techniques developed in the '30
SVD Algorithm

REAL PROBLEM:



We collect the real Dataset
 $\tilde{w}(t) = w(t) + \eta(t)$
 $\{\tilde{w}(0), \tilde{w}(1), \tilde{w}(2), \dots, \tilde{w}(N)\}$

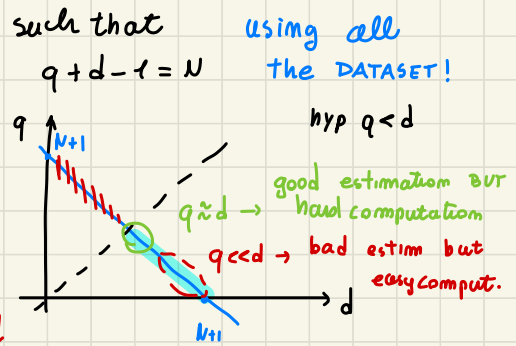
REAL 4SID ALG

1. Write $\tilde{H}_{qd}$ HANKEL MATRIX as

$[q \times d]$
(Rect Matrix)

$$\tilde{H}_{qd} = \begin{bmatrix} \tilde{w}(1) & \dots & \dots & \tilde{w}(d) \\ \vdots & & \ddots & \vdots \\ \tilde{w}(q) & \dots & \dots & \tilde{w}(q+d-1) \end{bmatrix}$$

d



2. factorize $\tilde{H}_{qd}$ thanks to the SVD Alg.

$[q \times d]$ $[q \times q]$ $[q \times d]$ $[d \times d]$

$$\tilde{H}_{qd} = \tilde{U} \tilde{S} \tilde{V}^T \quad \text{where } \tilde{U}, \tilde{V} \text{ are UNITARY MATRIX } \begin{cases} \det(M)=1 \\ M^T = M^{-1} \end{cases}$$

← SORT OF DIAGONALIZ. of Rect MATRIX

$$\tilde{S} = \begin{bmatrix} b_1 & & & 0 \\ & \ddots & & \\ & & b_q & \\ 0 & & & 0 \end{bmatrix} \quad \text{is the MATRIX of the sv of } \tilde{H}_{qd}, \text{ so:}$$
$$SV(M) = \sqrt{Eig(M M^T)} \quad \text{for a RECTANG MATRIX M}$$

3. identify SYSTEM ORDER and so divide the SYSTEM

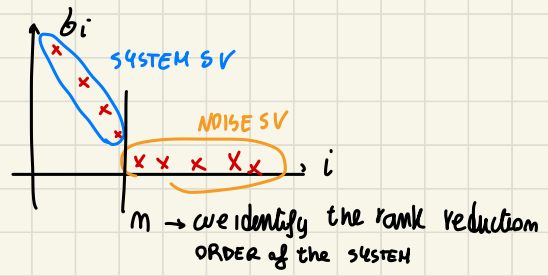
from $\tilde{S} = \begin{bmatrix} b_1 & & & \\ & \ddots & & \\ & & b_q & \\ & & & 0 \end{bmatrix}$ such that $b_1 \geq b_2 \geq \dots \geq b_q$

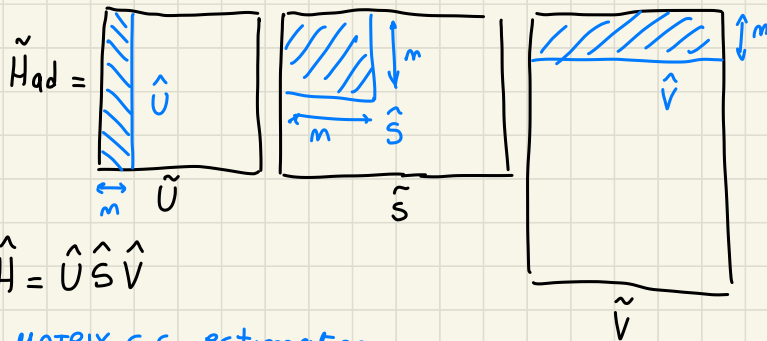
↓ SYSTEM

$$\tilde{H}_{qd} = \hat{H}_{qd} + \tilde{H}_{RESqd}$$

$\hat{H}_{qd}$ RANK $m \ll q$ HUGE RANK REDUCTION

$\tilde{H}_{RESqd}$ RESIDUAL, NOISE





4. MATRIX S.S estimation

$$\hat{H}_{qd} = (\hat{U} \hat{S}^{1/2}) (\hat{S}^{1/2} \hat{V}) = \hat{O}_{m+1} \cdot \hat{R}_{m+1}$$

$$\hat{H} = \hat{O}_{m+1} [1; :] \quad \hat{G} = \hat{R}_{m+1} [:; 1]$$

even IF the shift property
is STILL VALID and

→ we need to find

an APPROX LEAST SQUARE SOLUTION

$$O_2 = \underbrace{O_1}_{RECT} F \quad O_1^{-1} \nexists$$

In fact if you have a SYSTEM $Ax = B$
 $[h \times m][m \times 1] = [h \times 1]$

$$\hat{F} = (O_1^T O_1)^{-1} O_1^T O_2 = O_1^T O_2$$

$$\begin{cases} h < m & \text{more unk than eq} \rightarrow \infty \text{ SOL} \\ h = m & 1! \text{ SOL} \quad x = A^{-1}B \\ h > m & \text{LS SOL} \\ & \downarrow \\ & \tilde{x} = (A^T A)^{-1} A^T B \quad \text{LS SOL} \\ & \quad \quad \quad A^+ \text{ pseudo inverse} \end{cases}$$

NOTICE that LSID

can be extended for a general INPUT
 $u(z)$ applied

LSID OPTIMALITY? LSID Alg is OPTIMAL in the sense that we make the
 BEST rank reduction losing the least amount of information, so that our
 $\hat{H}_{resqd}$ is minimum, in fact in general we
 have ∞ possible RANK REDUCTION, we want the one that minimize the
 residual, OBTAINED THROUGH SVD Alg that guarantee minimal
 FORBENIUS NORM of $\hat{H}_{qdres}$

SVD has multiple usage

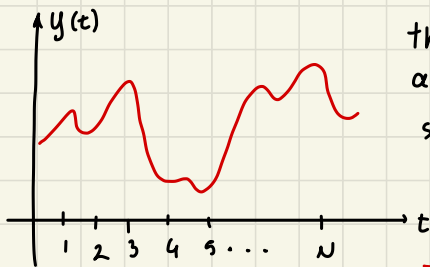
- complexity/order REDUCTION
- compression of information
- separation of INFO from NOISE

CH 2.1, END

Ch 2.2 B.B syst identification In time domain With PARAMETRIC approach in T.F description

We can estimate the model for 2 kind of systems { I/O
Time SERIES

TIME - SERIES



this can be a general signal

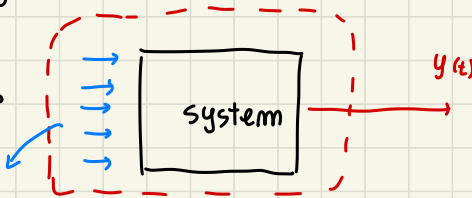
- NY company stock exchange
- points scored by a player
- PH10 in Milano

and also it depends on the $y(t)$ meaning

in general when dealing with Time series

MULTIPLE INPUTS.

- unmeasurable
- each with small contribute to $y(t)$

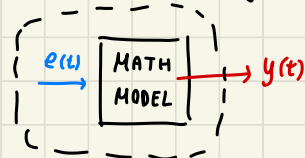


for EXAMPLE

rain
traffic
wind
⋮



We describe the system as



$e(t) \sim W N$ fictitious, un-measurable, Remote input

so we WORK with a DATASET

$\{y(1), y(2), \dots, y(N)\}$

as ONE FINITE Realization of a S.S.P

⇒ given a SSP We are interested on

• mean $\mu_y = E[y(t)]$

• covariance function $\gamma_y(\tau) = E[(y(t) - \mu_y)(y(t-\tau) - \mu_y)] \quad \tau \in \mathbb{Z}$

• spectral density $\left\{ \begin{array}{l} \Gamma_y(\omega) = \sum_{\tau=-\infty}^{+\infty} \gamma_y(\tau) e^{-j\omega\tau} \quad \omega \in \mathbb{R} \\ \gamma_y(\tau) = \frac{1}{2\pi} \int_{-\pi}^{+\pi} \Gamma_y(\omega) e^{j\omega\tau} d\omega \end{array} \right.$

DISCRETE FOURIER TRANSFORM

← same information

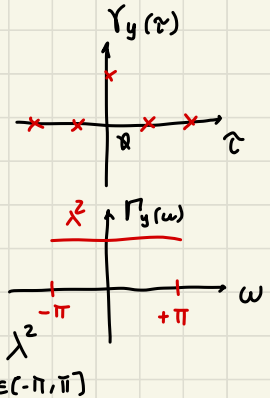
we can move between

• WHITE NOISE

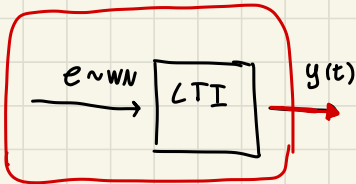
$e(t) \sim WN(0, \lambda^2)$ $\mu_e = 0$

COV. FUNCT $\left\{ \begin{array}{l} \gamma_y(0) = \lambda^2 \\ \gamma_y(\tau) = 0 \quad \forall \tau \neq 0 \end{array} \right.$

TOTALLY UN-PREDICTIBLE signal

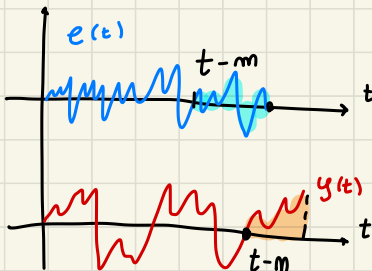


► We will model Time series as OUTPUT of a LINEAR discrete time dynamical system



DEFINITION a stochastic process $y(t)$ is said as "AUTO REGRESSIVE MOVING AVERAGE" of order m, m [ARMA(m, m)] if it has the following form

$$y(t) = \underbrace{a_1 y(t-1) + \dots + a_m y(t-m)}_{\text{AR}} + \underbrace{c_0 e(t) + \dots + c_m e(t-m)}_{\text{MA}}$$



depends on ● and ●

SPECIAL CASES:

$\left\{ \begin{array}{l} \text{AR}(m): \text{only simultaneous } e(t) \text{ effect} \\ \text{MA}(m): \text{NO RECURSIVE} \end{array} \right.$

Even if ARMA are a very general time series (S.P) description. We can have more complex processes requiring

M-ARMA: $y(t) = f((y(t-1), y(t-2), \dots, y(t-m), e(t), e(t-1), \dots, e(t-m)); \theta)$
NONLIN, STATIC, PARAMETRIC

When DESCRIBING **ARMA(m,m)** in T.F Domain

$$y(t) = \frac{C_0 + C_1 z^{-1} + \dots + C_m z^{-m}}{1 - a_1 z^{-1} - \dots - a_m z^{-m}} e(t)$$

$C(z) \rightarrow$ ITS ROOTS ARE THE ZEROS of ARMA
 $A(z) \rightarrow$ ITS ROOTS ARE THE POLES of ARMA

STATIONARITY

a S.P $y(t)$ is STATIONARY IF

- $m_y = E[y(t)] \quad \forall t$

- $\gamma_y(t_1, t_2) = \gamma_y(t_1 - t_2) = \gamma_y(\tau) \quad \forall t_1, t_2$
depends only on time shift

a RELEVANT RESULT told

that. IF $y(t)$ is OUTPUT of

the system $W(z)$, with input $v(t)$. IT is STATIONARY IF and only IF

- $v(t)$ is STATIONARY S.P
- $W(z)$ is Asymp stable

$\hookrightarrow$ so when dealing with ARMA

$e(t) \sim WN$ always s.s.p

STATIONARITY, GENERAL Results

We just need to check ARMA poles inside U.C

- AR(m):** $y(t) = \frac{z^m}{z^m - a_1 z^{m-1} - \dots - a_m} e(t)$
(IIR)

- m ZEROS in 0

- m GENERAL POLES

$\downarrow$
STATIONARITY depends on this

- MA(m):** $y(t) = \frac{C_0 z^m + C_1 z^{m-1} + \dots + C_m}{z^m} e(t)$
(F.I.R)

- m GENERAL ZEROS

- m POLES in 0

$\downarrow$
ALWAYS STATIONARY!

SPECTRUM

spectrum of a S.P im general

is computed as $\Gamma_y(\omega) = \sum_{\tau=-\infty}^{+\infty} \gamma_y(\tau) e^{-j\omega\tau}$

BUT, if $y(t)$ is the output of a system $W(z)$ asymp stable fed by

a S.S.P $v(t)$ so its spectrum is

$$\Gamma_y(\omega) = |W(e^{j\omega})|^2 \Gamma_v(\omega)$$

Where $W(e^{j\omega})$ is the

FREQUENCY RESPONSE of $W(z)$,

so it is the T.F evaluated on U.C

$$= |W(e^{j\omega})|^2 \lambda_c^2 \quad \text{IF } y \text{ is ARMA}$$

and so $v = e(t) \sim WN(0, \lambda_c^2)$

$|W(e^{j\omega})| \propto W(e^{j\omega})$ BODE mag^W , $W(e^{j\omega})$ NYQUIST Im- Re $\Gamma_c(\omega)$

ARMA PROCESS: ARMA(m,m)

• time domain $y(t) = a_1 y(t-1) + a_2 y(t-2) + \dots + a_m y(t-m) + c_0 e(t) + \dots + c_m e(t-m)$
with $e(t) \sim WN(0, \lambda^2)$

• T.F domain

$$y = \underbrace{[C(z)/A(z)]}_{W(z)} e(t)$$

• probabilistic domain

$$\begin{cases} \bullet my = \mathbb{E}[y(t)] \\ \bullet \gamma_y(\tau) = \mathbb{E}[y(t) \cdot y(t-\tau)] \quad \forall \tau \in \mathbb{Z} \end{cases}$$

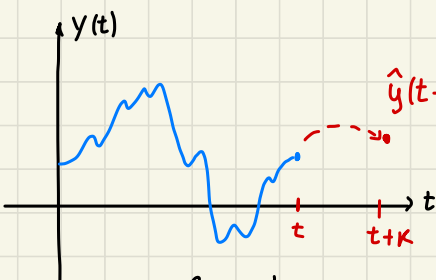
We can go from one to another representation

• spectral domain

$$\begin{cases} \bullet my = \mathbb{E}[y(t)] \\ \bullet \Gamma_y(\omega) = \sum_{\tau=-\infty}^{+\infty} \gamma(\tau) e^{-j\omega\tau} \end{cases}$$



PREDICTION OF ARMA



to solve it

We wanna find the OPTIMAL predictor from data

$$\{y(1), y(2), \dots, y(N)\}$$

$$\hat{y}(N+K|N)?$$

there are ∞ possible prediction definition, maybe using a weighted average of past samples...

↓ no one can find a better predictor than this!

- ① find the optimal model to fit the dataset (using the prediction theory! best model is the one which predict better)
- ② find optimal predictor from model obtained

⇓ We need preliminary definitions:

• ALL PASS FILTER $T(z)$ is a first order digital filter, defined by one parameter $a \in \mathbb{R}$ as:

$$T(z) = \frac{1}{a} \frac{1 + az^{-1}}{1 + \frac{1}{a}z^{-1}}$$

which has the relevant

features of having unitary norm $|T(e^{j\omega})|^2 = 1, \forall \omega, \forall a$

that guarantee that in the situation where $y(t) = T(z)v(t)$ with $v(t) \text{ s.s.p}$

follows $\Gamma_y(\omega) = |T(e^{j\omega})|^2 \Gamma_v(\omega) = \Gamma_y(\omega)$ y, v are spectrally equivalent
BUT NOT IDENTICAL because $T(e^{j\omega})$ introduce a phase shift on the output

• **CANONICAL FORM** Given a SSP modelled as an ARMA process

$$y(t) = (C(z)/A(z)) e(t) \quad e(t) \sim \text{WN}(0, \lambda^2)$$

we can say that this is the **CAN REP** of $y(t)$ if:

- C and A are same order
- C and A are co-prime
- C and A are monic
- C and A have all roots inside U.C

This is a particular unique ARMA representation usefull because it is the minimum order one
 ↳ without redundancies in the process definition!

(NOTE)

every ARMA(AR) process can always be transformed into a MA(∞) process:

for ex:

$$y(t) = \frac{1}{1 - \frac{1}{2}z^{-1}} e(t) \rightarrow y(t) = \frac{1}{2} y(t-1) + e(t) \quad \text{ARMA}(1,0)$$

$$\updownarrow$$

$$y(t) = \sum_{k=0}^{+\infty} \left(\frac{1}{2} z^{-1}\right)^k e(t) \rightarrow y(t) = e(t) + \frac{1}{2} e(t-1) + \dots \quad \text{MA}(\infty)$$

• **PREDICTION** $\Rightarrow$ Given a general ZERO-mean ARMA process in can form

$$y(t) = \frac{C(z)}{A(z)} e(t) \quad e(t) \sim \text{WN}(0, \lambda^2)$$

$$C(z) = 1 + c_1 z^{-1} + \dots + c_m z^{-m}$$

$$A(z) = 1 + a_1 z^{-1} + \dots + a_n z^{-n}$$

We wanna compute

$$\hat{y}(t+k|t)$$

DEFINING

$$\varepsilon(t+k) = y(t+k) - \hat{y}(t+k|t)$$

PREDICTION ERROR

$$y(t+k) = \hat{y}(t+k|t) + \varepsilon(t+k)$$

the predictor

is **OPTIMAL** y : $\mathbb{E}[\varepsilon(t+k) \hat{y}(t+k|t)] = 0$

meaning that $\varepsilon(t+k) = f(t+1, t+2, \dots, t+k)$ depends on future values

they must $\nLeftarrow \hat{y}(t+k|t) = f(t, t-1, t-2, \dots)$ depends on past

is UN-CORRELATED, so you have NO

important INFO in the error, all INFO on predictor!

to compute our predictor K -STEP ahead we perform long division K -STEP

$$\begin{array}{c|c} C(z) & A(z) \\ \vdots & E(z) \\ \hline \tilde{R}(z) & \end{array}$$

" $\tilde{R}(z) z^{-k}$ "

→ and from $C(z) = E(z)A(z) + R(z)$

↓ $C(z)/A(z) = E(z) + R(z)/A(z)$

$$y(t) = \underbrace{E(z) \cancel{e(t)}}_{\text{depends on } e(t), e(t-1), \dots, e(t-K+1)} + \underbrace{\frac{R(z)}{A(z)} e(t)}_{\text{depends on } e(t-K), e(t-K-1), \dots}$$

→ UN-CORRELATED
key $e(t)$ wwn

we neglect future values and build

$$\hat{y}(t|t-K) = \frac{R(z)}{A(z)} e(t) = \frac{\tilde{R}(z)}{\cancel{A(z)}} \cancel{e(t-K)}$$

PREDICTOR
from NOISE

→ unusefull!
you don't
KNOW $e(t)$!

OPTIMAL
because

$$E[\hat{y}(t|t-K) e(t)] = E\left[E(z) \cancel{e(t)} \cdot \frac{\tilde{R}(z)}{\cancel{A(z)}} \cancel{e(t-K)} \right] = 0$$

UNCORRELATED!

to find a usable predictor:

$$e(t) = \frac{A(z)}{C(z)} y(t) \quad \text{"WHITENING" FILTER}$$

predictor is a STATIONARY
signal having $C(z)$

← all roots inside U.C
(because CAN. FORM!)

$$\hat{y}(t|t-K) = \frac{R(z)}{C(z)} y(t) = \frac{\tilde{R}(z)}{C(z)} y(t-K)$$

↓
by definition

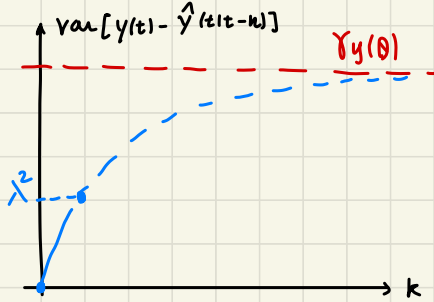
$$\varepsilon(t) = y(t) - \hat{y}(t|t-K) = E(z) e(t)$$

IT IS itself a S.S.P of type MA(K-1)

IN THE SPECIAL case of $K=1$

$$\begin{cases} \hat{y}(t|t-1) = \frac{C(z)-A(z)}{C(z)} y(t) \\ \varepsilon(t) = e(t) \end{cases}$$

PROPERTY of $\text{Var}[y(t) - \hat{y}(t|t-k)]$



- OBVIOUSLY: for $k=0$ it is 0
 - for $k=1$, $E(t) = e(t) \rightarrow \text{Var}[E(t)] = \lambda^2$
 - IT IS monotonically increasing with k
 - for $k \rightarrow \infty$ $\hat{y}(t|t-k) \rightarrow m_y$ (BEST one is mean)
- so $E(t) = y(t) - m_y \Rightarrow \text{Var}[E(t)] = \text{Var}[y(t)] = \gamma_y(0)$
- ↓ to define the QUALITY of the predictor we use the

error to
signal Ratio

$$\text{ESR}(k) = \frac{\text{Var}[E(t)]}{\text{Var}[y(t)]} \in [0, 1]$$

↑
OPTIMAL

WORST POSSIBLE

Sometimes even the optimal predictor does NOT guarantee a good ESR because the process can be intrinsically hard to predict!

notice that $\hat{y}(t|0) \simeq m_y$ when we need to estimate it

ARIMA it is NOT always possible to find the equivalent can. representation. IF in fact a S.S.P has ZERO/POLES in the u.c they cannot be cancelled!

in particular, when $A(z)$ has ROOTS in $+1$, (INTEGRATOR)

$$y(t) = \frac{C(z)}{A(z)(z-1)^d} e(t) \quad e \sim \text{WN}(0, \lambda^2)$$

→ **ARIMA**(m, d, m)

particular case when $d=1 \Rightarrow y(t) = \frac{z}{z-1} e(t)$ **ARIMA**(0,1,0)

$$y(t) = y(t-1) + e(t) \quad \text{"RANDOM WALK"} \quad \text{NOT STATIONARY}$$

IDENTIFICATION of ARMA MODELS from Data

We use a classical parametric approach...

STEP 1) Data collection and pre-processing $\{y(1), y(2), \dots, y(N)\}$

STEP 2) prior choice of a parametric class of math models $m(\theta)$

In our case ARMA:

$$m(\theta): y(t) = \frac{1 + c_1 z^{-1} + \dots + c_m z^{-m}}{1 + a_1 z^{-1} + \dots + a_m z^{-m}} e(t) \quad e \sim WN(0, \sigma^2)$$

$$\theta = (a_1 \ a_2 \ \dots \ a_m \ c_1 \ c_2 \ \dots \ c_m)^T \quad \theta \in \mathbb{R}^{m+m} \quad m_d = m+m$$

STEP 3) performance index definition $J(\theta) \geq 0$, $J(\theta): \mathbb{R}^m \rightarrow \mathbb{R}^+$
 which is a qualitative measurement of the quality of the model
 $m(\theta_1)$ is better than $m(\theta_2)$ if $J(\theta_1) < J(\theta_2)$

$$J_N(\theta) = \frac{1}{N} \sum_{t=1}^N (y(t) - \hat{y}(t|t-1; \theta))^2 \quad (\text{P.E.M})$$

this is the sample variance
 of the prediction error
 $\approx \chi^2(\theta)$

STEP 4) $J(\theta)$ OPTIMIZATION

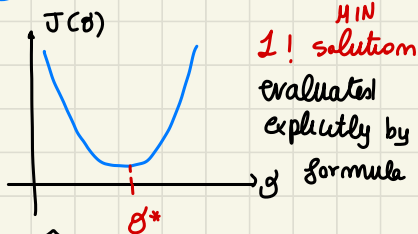
$$\rightarrow \theta^* = \arg \min_{\theta} \{J_N(\theta)\} \rightarrow m(\theta^*) \text{ BEST MODEL}$$

STEP 5) VALIDATION

in the OPTIMIZATION

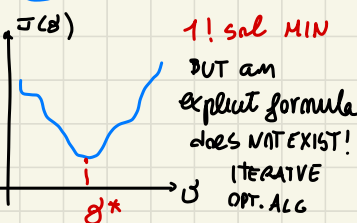
procedure there can be 3 cases

① $J(\theta)$ is quadratic on θ



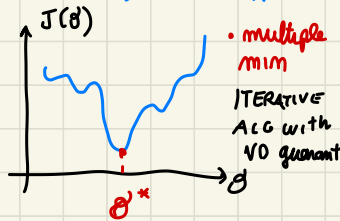
↑
 per index in a AR(m)
 identification setting

② $J(\theta)$ JUST CONVEX



↑
 $J(\theta)$ when
 dealing with general
 ARMA(m,m) process id!

③ $J(\theta)$ NOT CONVEX



IDENTIFICATION of AR

1) DATA collection $\{y(1), y(2), \dots, y(N)\}$

2) model selection

$$m(\theta): y(t) = \frac{1}{A(z)} e(t) \quad e \sim WN(0, \lambda^2) \quad A(z) = 1 + a_1 z^{-1} + \dots + a_m z^{-m}$$

$$\theta = (a_1 \dots a_m)^T \quad m\theta = m$$

3)

$$J_N(\theta) = \frac{1}{N} \sum_{t=1}^N \left(y(t) - \hat{y}(t|t-1) \right)^2 = \hat{y}(t|t-1) = -a_1 y(t-1) - a_2 y(t-2) - \dots - a_m y(t-m)$$

$$= \frac{1}{N} \sum_{t=1}^N \left(y(t) - \varphi^T \theta \right)^2 \quad \text{QUADRATIC in } \theta, \text{ easy to sol}$$

$$\varphi = \begin{pmatrix} -y(t-1) \\ -y(t-2) \\ \vdots \\ -y(t-m) \end{pmatrix}$$

4)

$$\frac{\partial J_N}{\partial \theta} = -\frac{2}{N} \sum_{t=1}^N \varphi(t) (y(t) - \varphi^T(t) \theta) = 0$$

$$\Rightarrow \theta^* = \left(\sum_{t=1}^N \varphi(t) \varphi^T(t) \right)^{-1} \left(\sum_{t=1}^N \varphi(t) y(t) \right) \quad \text{LS Formula Unique solution}$$

IF we

assume predict

$$\lambda_e^2 \approx J(\theta^*)$$

IDENTIFICATION of ARMA

1) collect the DATASET $\{y(1), y(2), \dots, y(N)\}$

2) selection of model class

$$m(\theta): y(t) = \frac{C(z; \theta)}{A(z; \theta)} e(t) \quad e \sim WN(0, \lambda^2)$$

$$m\theta = m+m$$

$$\theta = (a_1 a_2 \dots a_m \quad c_1 c_2 \dots c_m)^T$$



$$C(z; \theta) = 1 + c_1 z^{-1} + \dots + c_m z^{-m}$$

$$A(z; \theta) = 1 + a_1 z^{-1} + \dots + a_m z^{-m}$$

3) definition of

$$J_N(\theta) = \frac{1}{N} \sum_{t=1}^N \left(y(t) - \frac{C(z; \theta) - A(z; \theta)}{C(z; \theta)} y(t) \right) = \frac{1}{N} \sum_{t=1}^N \frac{A(z; \theta)}{C(z; \theta)} y(t)$$

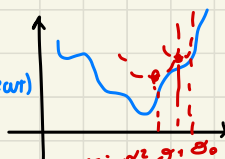
3rd KIND OF OPTIMIZ... NOT explicit

4) $\theta^* = \underset{\theta}{\operatorname{argmin}} \{J_N(\theta)\}$ MINIMIZE of

$J_N(\theta)$ using an ITERATIVE (OR QUASI-linear)

Algorithm $\rightarrow$ Newton method

(True can exist local min $\rightarrow$ RE COMPUTE)



$$\theta^{(0)} \rightarrow \theta^{(1)} \rightarrow \dots \rightarrow \theta^*$$

try to converge to BEST solution,

RE ITERATE =

NO GUARANTEE

CHOICE of the ORDER of the MODEL

how to select properly $m(\theta)$? We try ALL choices and chose the BEST of ALL
 selecting a set of BALANCED ARMA process

ARMA (1,0) $m_\theta = 1$

ARMA (1,1) $m_\theta = 2$

ARMA (2,1)

ARMA (2,2)

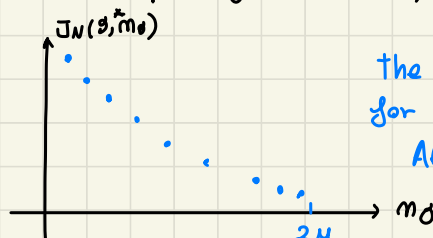
$\vdots$

ARMA (M,M)

$m_\theta = 2M$

So our decisional variable is JUST m_θ
 1 dim optimiz. problem

$\Rightarrow$ and we find the OPTIMAL model $\forall m_\theta$
 than plotting $J_N(\theta, m_\theta)$



the BEST $\rightarrow$
 for $m_\theta = 2M$

Always..

MORE complex
 fits better!

We need a **Trade off**

between a good fitting
 and generalization

ability of our model $\Rightarrow$

We split DATASET

CROSS VALIDATION

$\{y(1), y(2), \dots, y(N/2), y(N/2+1), \dots, y(N)\}$

ϕ_T TRAINING

ϕ_V VALIDATION

• select m_θ

• compute optimal model

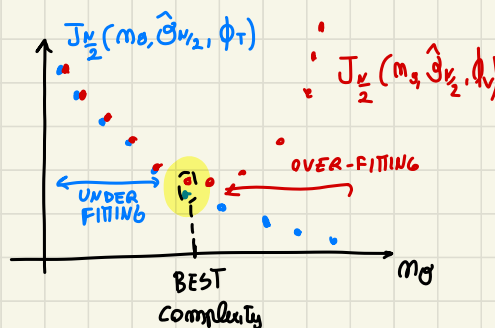
minimizing $J_{N/2}(\theta, m_\theta, \phi_T) \rightarrow \theta_{N/2}^*$

• compute $J_{N/2}(\hat{\theta}_{N/2}, m_\theta, \phi_T)$

$J_{N/2}(\hat{\theta}_{N/2}, m_\theta, \phi_V)$

• repeat $\forall m_\theta = 1, \dots, 2M$ (ITERATE $\forall m_\theta$)

$\Rightarrow$



once found the overall BEST model

$m(\theta)$, we evaluate $\varepsilon(t) = y(t) - \hat{y}(t|t-1; \theta^*)$ $t=1, 2, \dots, N$

and the model is OPTIMAL if

$\varepsilon(t) \sim WN$

WHITENING TEST! meaning
 that we do the BEST

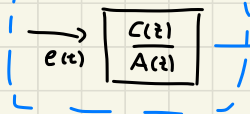
IF NOT... maybe "LINEAR" ARMA is NOT enough,

we should use NARMA models for a good identification

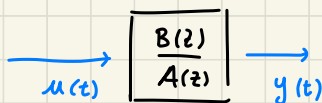
IDENTIFICATION / PREDICTION OF I/O SYSTEMS

until now we work with systems modelled as

time-series



What about I/O



Where we can measure / access $\{u(1), u(2), \dots, u(N)\}$

$\{y(1), y(2), \dots, y(N)\}$

We repeat all the

theory done on time series for I/O syst!

IF we use same approach

1) DATASET

collection + learning

$\{u(1), u(2), \dots, u(N)\}$

$\{y(1), y(2), \dots, y(N)\}$

design the experiment!

2) model selection

$$m(\theta): y(t) = \frac{B(z)}{A(z)} z^{-1} u(t)$$

$$B(z) = b_0 + b_1 z^{-1} + \dots + b_p z^{-p}$$

$$A(z) = 1 + a_1 z^{-1} + \dots + a_m z^{-m}$$

$$\theta = (a_1 \dots a_m \ b_0 \dots b_p)^T$$

$$m_\theta = m + p + 1$$

3) P.E.M $J_N(\theta)$

here the predictor is simply

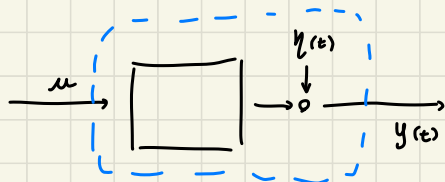
$$\hat{y}(t) = \hat{y}(t|t-1) = -a_1 \hat{y}(t-1) - a_2 \hat{y}(t-2) - \dots - a_m \hat{y}(t-m) + b_0 u(t-1) + \dots + b_p u(t-p-1)$$

$$\text{so } J_N(\theta) = \frac{1}{N} \sum_{t=1}^N (y(t) - \hat{y}(t|t-1; \theta))^2$$

QUADRATIC
in θ

$$4) \theta^* = \arg \min_{\theta} \{J_N(\theta)\} \rightarrow m(\theta^*)$$

BAD MODEL, WRONG model id. The issue is that we are COMPLETELY NEGLECTING the real behaviour of the system
We have to include NOISE in the model structure!



new class
of STOCHASTIC
PROCESSES

a S.P $y(t)$ is **ARMAX** $(m, m, p+k)$, if it has the following form:

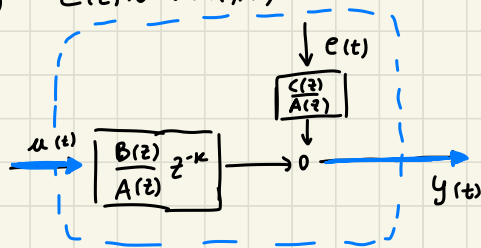
$$y(t) = \underbrace{a_1 y(t-1) + \dots + a_m y(t-m)}_{\text{AR } (m)} + \underbrace{c_0 e(t) + c_1 e(t-1) + \dots + c_m e(t-m)}_{\text{MA } (m)} + \underbrace{b_0 u(t-k) + \dots + b_p u(t-k-p)}_{\times (p+k)}$$

↓

$$y(t) = \frac{C(z)}{A(z)} e(t) + \frac{B(z)}{A(z)} u(t-k) \quad e(t) \sim WN(0, \lambda^2)$$

PREDICTION OF ARMAX

once described the model
we do the same reasoning
done on ARMA, perform
KSTEP LONG DIVISION and obtain



$$y(t) = \underbrace{E(z)}_{\text{UN-PREDICT}} e(t) + \frac{R(z)}{A(z)} e(t) + \frac{B(z)}{A(z)} u(t-k) = \hat{y}(t|t-k) \quad \text{PREDICTOR from NOISE (un-usefull!)}$$

$$E(t) = E(z)e(t)$$

by inverting the dynamic $e(t) = \frac{A(z)}{C(z)} y(t) - \frac{B(z)}{A(z)} u(t-k)$

We plug-in $e(t)$ on

the predictor from noise and find out

$$\hat{y}(t|t-k) = \frac{R(z)}{C(z)} y(t) + \frac{B(z)E(z)}{C(z)} u(t-k)$$

still

$$E(t) = E(z)e(t) \quad \text{the exogenous part does NOT introduce error!}$$

special case $k=1$

$$\begin{cases} \hat{y}(t|t-1) = \left(\frac{C(z)-A(z)}{C(z)} \right) y(t) + \frac{B(z)}{C(z)} u(t-1) \\ E(t) = e(t) \end{cases}$$

IDENTIFICATION of ARMAX from DATA

1) Dataset collection $\{u(1), u(2), \dots, u(N)\}$, $\{y(1), y(2), \dots, y(N)\}$
(DESIGN EXPERIMENT)

2) model class definition

$$m(\theta): y(t) = \underbrace{\frac{C(z; \theta)}{A(z; \theta)}}_{\text{CAN FORN}} e(t) + \frac{B(z; \theta)}{A(z; \theta)} u(t-1) \quad \begin{matrix} \rightarrow \theta = (a_1 \dots a_m \ c_1 \dots c_m \ b_0 \dots b_p)^T \\ m_\theta = m+m+1+p \\ e(t) \sim WN(0, \lambda^2) \end{matrix}$$

even if we choose $k=1$
we don't lose generality of
the model, the identification
of non zero $\hat{b}_i$ tell us the delay

3) P.E.M to define per. index

$$J_N(\theta) = \frac{1}{N} \sum_{t=1}^N \left(y(t) - \frac{C(z; \theta) - A(z; \theta)}{C(z; \theta)} y(t) - \frac{B(z; \theta)}{C(z; \theta)} u(t-1) \right)^2 =$$

$$= \frac{1}{N} \sum_{t=1}^N \left(\frac{A(z; \theta)}{C(z; \theta)} y(t) - \frac{B(z; \theta)}{C(z; \theta)} u(t-1) \right)^2$$

4) optimization $\theta^* = \arg \min \{J_N(\theta)\}$

in case of ARX(m, p+1)

- J_N QUADRATIC on θ for ARX
- J_N complex to optimize for ARMAX, MAX

We have $C(z; \theta) = 1$ NOT θ @ den of $J_N(\theta)$

easy to solve explicitly

$$\theta^* = \left[\sum_{t=1}^N \varphi(t) \varphi(t)^T \right]^{-1} \left[\sum_{t=1}^N y(t) \varphi(t) \right]$$

• the best order of model m, m, p is chosen by using CROSS VALID.

• to see if the model is optimal we perform a

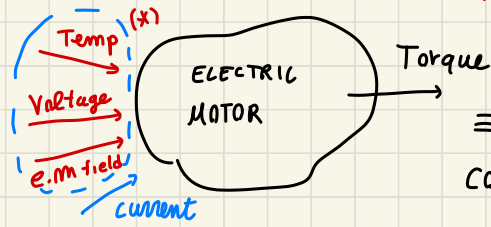
Whitening TEST on $\varepsilon(t; \theta^*) = y(t) - \hat{y}(t|t-1; \theta^*) \quad t=1 \dots N$

↓

IF $\varepsilon(t)$ NOT WN $\Rightarrow$ may LINEAR is NOT enough!

ARMA vs ARMAX (HOW TO CHOOSE)

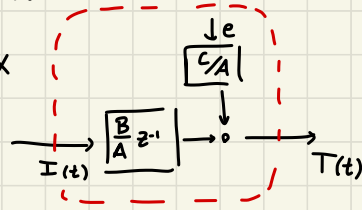
Example 1 motor dynamic



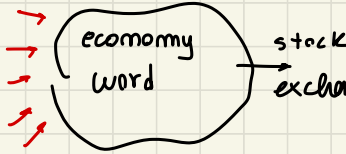
(*) Input signal u_m - meas, or JUST affecting less the Torque!

$\equiv$ ARMAX

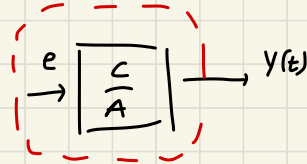
CONTROLLABLE



Example 2 economy



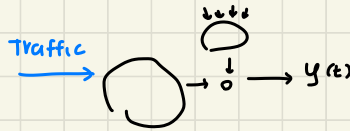
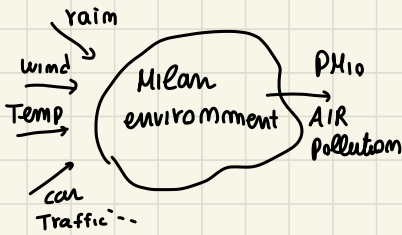
$\equiv$ ARMA



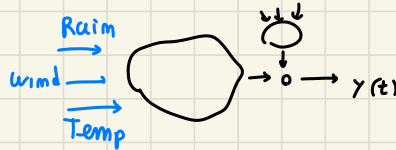
Example 3 HARD choice

- it is a control var?
- are in measurable?
- Are there dominant INPUT?

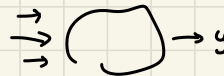
(CHOICE 1) ARMAX CONTROLLABLE



(CHOICE 2) ARMAX MEASURABLE



(CHOICE 3) ARMA



COMPARISON

ARMA

- simple to tune (y(t)) ✓
- minimum predictor quality ✗

ARMAX

- good predictor quality ✓
- can be used for control ✓
- hard to tune complex ✗

Ch 2.3 B.B I/O syst. identification IM T.F domain

with PARAMETRIC approach im FREQ. domain

We use a classical Parametric approach with some particularity in the design of experiment and definition of perf. Index

● relevant steps

step 1) Experiment Design and Data collection / pre-cleaning

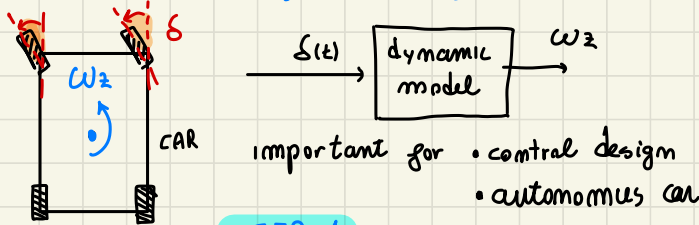
step 2) parametric model selection $m(\Theta)$

step 3) performance index definition $J(\Theta)$

step 4) optimization $\Theta^* = \arg \min_{\Theta} \{J(\Theta)\} \rightarrow m(\Theta^*)$

KEY IDEA: make a set of experiments on a certain frequency range, $\forall$ estimate $W(e^{j\omega})$ points $\forall$ experiment, fit the measured $W(e^{j\omega})$ with the model

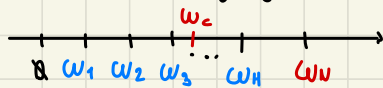
EXAMPLE steer to yaw-rate dynamic



STEP 1

EXPERIMENT DESIGN

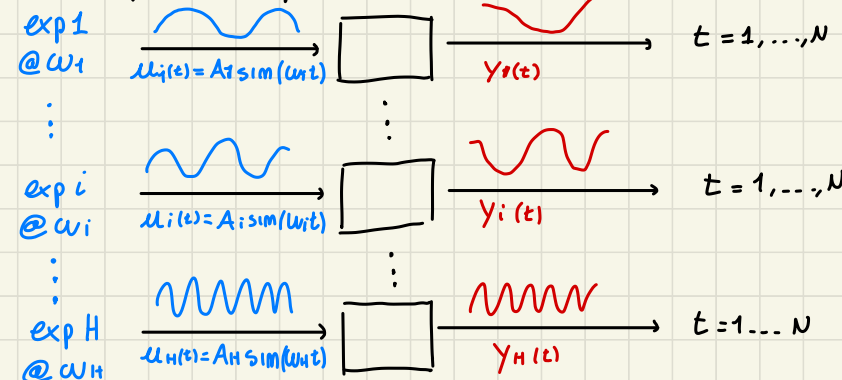
We select a range of relevant frequencies, and a set of evenly spaced frequency



such that

$$\omega_c < \omega_H < \omega_N$$

and we make H independent experiments (to catch all the important dynamical features)



Focus on general experiment i

INPUT AMPLITUDE you must take care on the actuator power limit

for example, on a steer dynamic

$$T(t) \approx K \delta(t)$$

$$\text{IF } \delta(t) = u(t) = A_i \sin(\omega_i t)$$

$$P(t) = T(t) \cdot \dot{\delta}(t) \text{ steering power}$$

$$P(t) = \underbrace{A_i^2 k \omega_i}_{\text{POWER amplitude}} \sin(\omega_i t) \cos(\omega_i t)$$

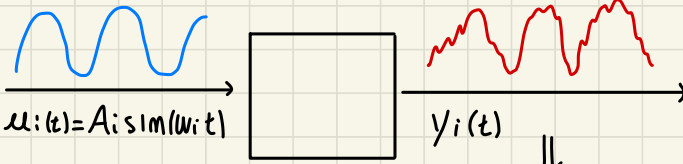
we can request $\bar{P} = \text{constant}$, so

$$A_i = \sqrt{\frac{\text{constant}}{k \omega_i}}$$

$$A_i \approx \frac{1}{\sqrt{\omega_i}}$$

for increasing freq we decrease amplitude of excitement to respect power actuation limit!

OUTPUT



model identification problem

we wanna clean the output measurement from noise!

$$y_i(t) = B_i \sin(\omega_i t + \varphi_i)$$

ideally for freq resp Th

↓
IN REALITY we have NOISE

- on measurement
- on the syst + NON LIN.

We decide to fit the measurements $y_i(t)$ with ideal model

$$\hat{y}_i(t) = B_i \sin(\omega_i t + \varphi_i) = a_i \sin(\omega_i t) + b_i \cos(\omega_i t)$$

use this model for fitting because it is linear on the unknowns

$$J_N(a_i, b_i) = \frac{1}{N} \sum_{t=1}^N (y_i(t) - a_i \sin(\omega_i t) - b_i \cos(\omega_i t))^2$$

$$\frac{\partial J_N}{\partial a_i} = 0 \Rightarrow (\hat{a}_i, \hat{b}_i) \text{ optimal model}$$

$$\frac{\partial J_N}{\partial b_i} = 0 \Rightarrow \text{then from here you come back to } (\hat{B}_i, \hat{\varphi}_i) \text{ description (which is more intuitive)}$$

getting the output

measurement cleaned as

$$y_i(t) = \hat{B}_i \sin(\omega_i t + \hat{\varphi}_i)$$

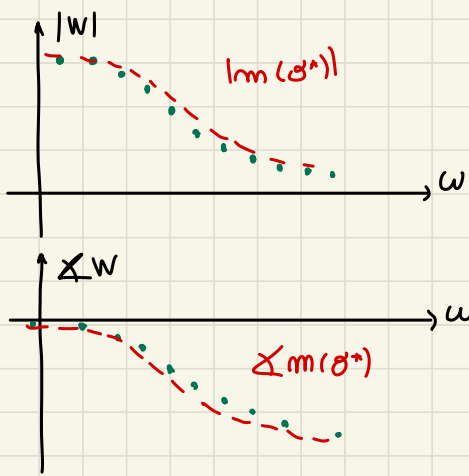
output of system feed by

$$u_i(t) = A_i \sin(\omega_i t)$$

⇒ you can compute the sample freq resp. as

$$W(e^{j\omega_i}) \approx \frac{\hat{B}_i}{A_i} e^{j\hat{\varphi}_i}$$

on repeat $\forall i=1, \dots, H$ ($\forall$ experiment)



$$\left(\frac{\hat{B}_i}{A_i} e^{j\hat{\varphi}_i} \right)$$

STEP 2 we select a simple classical model structure to fit

$$W(z, \theta) = \frac{b_0 + b_1 z^{-1} + \dots + b_p z^{-p}}{1 + a_1 z^{-1} + \dots + a_m z^{-m}} z^{-1}$$

$$\theta = (b_0 \dots b_p \ a_1 \dots a_m)^T$$

STEP 3 define a perf index on freq domain:

$$J_H(\theta) = \frac{1}{H} \sum_{i=1}^H \left| \frac{\hat{B}_i}{A_i} e^{j\hat{\varphi}_i} - W(e^{j\omega_i}; \theta) \right|^2$$

STEP 4

optimization: $\theta^* = \underset{\theta}{\operatorname{argmin}} \{ J_H(\theta) \} \rightarrow m(\theta^*)$

usually an HARD optimiz problem with cost function NOT CONVEX- NOT QUADRATIC

Freq Domain Id:

- Robust to Noise, Reliable (High signal to Noise Ratio)
- Intuitive identification
- consistent with control Theory (Loop shaping on freqdom.)

- complex experiment design
- NO noise model at the end (we clean all)

(Notice that this method can also be applied using different kind of input on the system)

Ch 2. 4 MODEL BASED SW SENSING, USING K.F

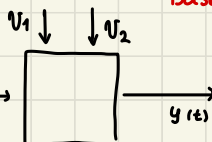
Based on a s.s description

$$\begin{cases} x(t+1) = Fx(t) + Gu(t) + Jv_1(t) \\ y(t) = Hx(t) + Dv_2(t) + v_2(t) \end{cases}$$

system description
found with white

box techniques (using physical analysis of syst.)

NOISES included



having as DATASET

$\{u(1), u(2), \dots, u(t)\}$

$\{y(1), y(2), \dots, y(t)\}$

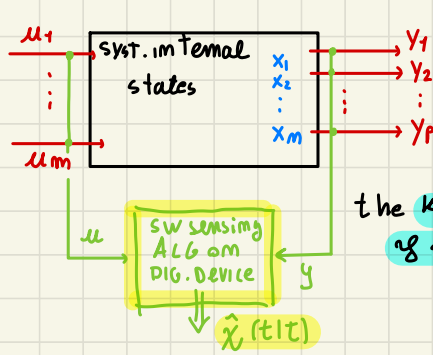
KALMAN FILTER

goals

- k step ahead output prediction $\hat{y}(t+k|t)$
- k step ahead STATE prediction $\hat{x}(t+k|t)$
- STATE estimation / filtering $\hat{x}(t|t)$ → Relevant to estimate un-measurable signals ($\hat{y}(t|t)$ is meaningless!)

Relevant to estimate un-measurable signals

usually in a system



MIMO

M INPUT

P OUTPUT (measured)

M STATES

$\{m \gg p\}$

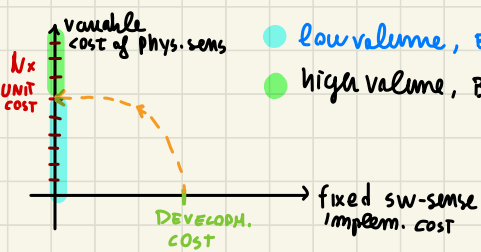
the KNOWLEDGE

of STATES is important for

- CONTROL
- MONITORING

SW Sensing VS PHYSICAL SENSORS

- necessary for un-measurable variables
- sometimes IT is cheaper



- low volume, BETTER physical
- high volume, Better SW-sense

KEY QUESTION

- cheaper?
- feasible?
- accurate?

When developing SW sensors

usage for example:

- sometimes we use both SW + phys sensors for redundancy!

SLIP ESTIMATION

$$\lambda = \frac{cur - v}{v}$$

NOT 100% AVAILABLE by phys sensors!

BASIC K.F

{ 1 STEP PREDICTION }



SYST DESCRIPTION

$$\begin{cases} x(t+1) = Fx(t) + Gu(t) + v_1(t) \\ y(t) = Hx(t) + v_2(t) \end{cases}$$

⇒ EXTENSIONS

- FILTER ($K=0$)
- K-STEP PREDICTION
- EXT. INPUT
- NON LINEAR syst (EKF)
- TIME-VARYING

$$x(t) = \begin{pmatrix} x_1(t) \\ \vdots \\ x_m(t) \end{pmatrix} \quad y(t) = \begin{pmatrix} y_1(t) \\ \vdots \\ y_p(t) \end{pmatrix} \quad \mu(t) = \begin{pmatrix} \mu_1(t) \\ \vdots \\ \mu_m(t) \end{pmatrix}$$

NOISES DESCRIPTION

• $v_1(t) = \begin{pmatrix} v_{11}(t) \\ \vdots \\ v_{1m}(t) \end{pmatrix}$ STATE/MODEL NOISE $v_1(t) \sim WN(0, V_1)$

$$\begin{cases} \mathbb{E}[v_1(t)] = (0 \ 0 \dots 0)^T \\ \mathbb{E}[v_1(t) \cdot v_1^T(t)] = V_1 \quad V_1 \geq 0 \text{ spd COVARIANCE MATRIX} \\ \mathbb{E}[v_1(t) \cdot v_1(t-\tau)] = 0 \quad \forall t, \tau \neq 0 \text{ Whiteness} \end{cases}$$

• $v_2(t) = \begin{pmatrix} v_{21}(t) \\ \vdots \\ v_{2p}(t) \end{pmatrix}$ OUTPUT/MEASUREMENT NOISE $v_2(t) \sim WN(0, V_2)$

$$\begin{cases} \mathbb{E}[v_2(t)] = (0 \ 0 \dots 0)^T \\ \mathbb{E}[v_2(t) \cdot v_2^T(t)] = V_2 \quad V_2 > 0 \text{ (p.d.) COV. MATRIX} \\ \mathbb{E}[v_2(t) \cdot v_2(t-\tau)] = 0 \quad \forall t, \tau \neq 0 \text{ Whiteness} \end{cases}$$

hyp. for K.F existence

Hyp: $\mathbb{E}[v_1(t) \cdot v_2(t-\tau)] = \begin{cases} 0 & \forall \tau \neq 0 \\ \underline{\widehat{V}_{12}} & \tau = 0 \end{cases}$ (usually $V_{12} = 0$, NOT ALWAYS)

(m x p) CROSS-VARIANCE MATRIX

considering the initial condition of

$$\mathbb{E}[x(1)] = x_0$$

$$\mathbb{E}[(x(1) - x_0)(x(1) - x_0)^T] = p_0$$

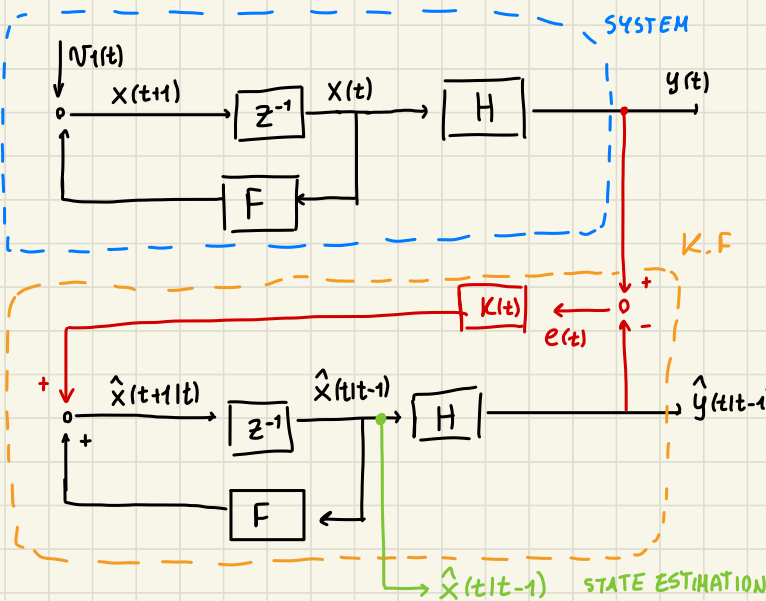
$$\text{and } \begin{cases} x(1) \perp v_1(t) \\ x(1) \perp v_2(t) \end{cases}$$

↳ we can find the K.F expression

K.F SOLUTION (BASIC)

$$\text{I.C } \begin{cases} \hat{x}(1|0) = \mathbb{E}(x(1)) = x_0 \\ P(1) = P_0 = \text{var}[x(1)] \end{cases}$$

$$\begin{cases} \hat{x}(t+1|t) = F \hat{x}(t|t-1) + K(t) \cdot e(t) & \text{STATE eq} \\ \hat{y}(t|t-1) = H \hat{x}(t|t-1) & \text{OUTPUT eq} \\ e(t) = y(t) - \hat{y}(t|t-1) & \text{PREDICTION ERROR} \\ K(t) = (F P(t) H^T + V_{12}) (H P(t) H^T + V_2)^{-1} & \text{K.F GAIN} \\ P(t+1) = (F P(t) F^T + V_1) - (F P(t) H^T + V_{12}) (H P(t) H^T + V_2)^{-1} (F P(t) H^T + V_{12})^T & \text{DIFF RICCATI EQ (DRE)} \end{cases}$$



IDEA :

- make a simulated replica of the system (thanks to F, H, knowledge)
- compare real output with estimated one
- make a feedback correction on the replica system

Comments

- DRE is a difference nonlinear equation, Autonomous dynamical syst eq

$$\begin{cases} P(t+1) = f(P(t)) \\ P(1) = P_0 \end{cases}$$

- We are sure that K.F has a solution because $(\underbrace{H P(t) H^T + V_2}_{\text{spd}})$ is p.d
 $\underbrace{\text{pd by Hyp.}}$

- K(t) and P(t+1) has a standard formulation respect blocks

state block $(F P F^T + V_1)$

output block $(H P H^T + V_2)$

Mix block $(F P H^T + V_{12})$

$$(\square P(t) \diamond + \text{Noise})$$

$$\Rightarrow P(t+1) = (\text{STATE}) - (\text{MIX})(\text{OUTPUT})^{-1}(\text{MIX})^T$$

$$K(t) = (\text{MIX}) \cdot (\text{OUTPUT})^{-1}$$

$$P(t) = \mathbb{E}[(x(t) - \hat{x}(t|t-1))(x(t) - \hat{x}(t|t-1))^T] = \text{var}[x(t) - \hat{x}(t|t-1)]$$

$P(t)$ is the state prediction error variance (1 step ahead)

- even if at first view K.F seems just a construction procedure using white box S.S system model, in Reality there is an issue of identification of V_1, V_2, V_3 that can be estimated empirically $\rightarrow$ so in reality we need a DATASET! (ML procedure)

- the idea behind K.F structure is the "STATE OBSERVER"

so to make a model twin, KALMAN develop the explicit GAIN $K(t)$ formula

$K(t)$ as $[M \times P]$ MATRIX GAIN (hard to tune empirically! $\rightarrow$ [small K : poor perform. HIGH K : UNSTABILITY ISSUE])

K.F EXTENSIONS

1 MULTI-STEP (K) PREDICTOR

once found $\hat{x}(t+1|t)$ the best to predict $\hat{x}(t+k|t)$ is to formulate the new predictor as

$$\begin{cases} \hat{x}(t+k|t) = F^{k-1} \hat{x}(t+1|t) \\ \hat{y}(t+k|t) = H \hat{x}(t+k|t) \end{cases}$$

2 FILTER

Reasoning as before, if F is invertible: $\hat{x}(t|t) = F^{-1} \hat{x}(t+1|t)$

BUT IF $F^{-1} \nexists$

with ISSUE of INVERSION, we have the K.F IM FILTER FORM

WITH $V_2 = 0$ HYP!

$$\begin{cases} \hat{x}(t|t) = F \hat{x}(t-1|t-1) + K_0(t) e(t) \\ \hat{y}(t|t) = H \hat{x}(t|t-1) \\ e(t) = y(t) - \hat{y}(t|t-1) \\ K_0(t) = (P(t)H^T)(HP(t)H^T + V_2)^{-1} \\ P(t+1) = (FP(t)F^T + V_1) - (FP(t)H^T)(HP(t)H^T + V_2)^{-1}(FP(t)H^T)^T \end{cases}$$

3 EXTERNAL INPUT

being $G u(t)$ an exogenous deterministic part, it does NOT affect $e(t)$, so neither the DRE $P(t+1)$...

all K.F structure remain the same, including also the G block on the structure

($P(t)$ is the var of prediction state error $\rightarrow$ same!)

4 TIME VARYING SYSTEM When the system is described by matrix

$F(t), H(t), G(t) \rightarrow$ we simply extend K.F by using the same K.F formula but with time dependent matrix

K.F ready for LTV systems

5 NON LINEAR SYSTEMS (end of ch 2.4, LATER!) $\Rightarrow$ EXTENDED K.F (EKF)

ASYMPTOTIC K.F the issue of K.F is that even if the initial system is LTI because of the updating K. Gain $K(t)$, the overall K.F comes a LTV system

LTV K.F issues

- Asympt stability guarantee difficult to check! In LTV systems, if $F(t)$ is the STATE MATRIX, even having $\text{all } |\text{Eig}(F(t))| < 1$ Does NOT guarantee!
- can be computationally demanding (more for high order syst) to evaluate $K(t), P(t+1)$ at each time step!

$\Downarrow$ we wanna find a LTI K.F solution

IF $P(t+1) \rightarrow \bar{P}$ for $t \rightarrow \infty$

equivalent $K(t) \rightarrow \bar{K}$ we can use a constant gain system and assuming existence and convergence:

STABILITY

$$\hat{x}(t+1|t) = F \hat{x}(t|t-1) + \bar{K} (y(t) - H \hat{x}(t|t-1)) + \underbrace{\bar{G} u(t)}_{\text{NOT necessary for this analysis}}$$

$$\hat{x}(t+1|t) = (F - \bar{K}H) \hat{x}(t|t-1) + \bar{K} y(t) + G u(t)$$

equivalent

STATE MATRIX

$\Rightarrow$ stability of Asymp K.F related to

(you can have unstable

system F , but stable K.F)

All $\text{Eig}(F - \bar{K}H)$ inside U.C

EXISTENCE

$\bar{K}$ exists if $\bar{P}$ exist, with $\bar{P}$ steady-state solution of DRE, so they are equilibrium position of DRE Autonomous non lin. dyn syst

$$f(\bar{x}) = \bar{x}$$

DISCRETE TIME
STEADY SOL.

$$\rightarrow \bar{P} = (F \bar{P} F^T + V_1) - (F \bar{P} H^T + V_2) (H \bar{P} H^T + V_2)^{-1} (F \bar{P} H^T + V_2)^T$$

ALGEBRAIC RICCATI Eq (ARE)

(steady STATE solution of DRE)

IF we find $\bar{P}$... we wanna be sure that

- **Existence** ARE has a spd solution? (1)
- **convergence** if there exist a spd sol, we assume that $P(t) \rightarrow \bar{P} \quad \forall P_0$ init. cond (2)
- **stability** if EXIST and CONV guaranteed, we want that the CORRESPONDING $\bar{K}$ is such that $(F - \bar{K}H)$ has all Eig inside U.C (3)

↑

this is hard to verify by hand (more when high dim syst),
BUT luckily we have some relevant Result to guarantee this properties

as SUFFICIENT CONDITION for (1),(2),(3) → **ASYMPTOTIC K.F Theorems**

↓
very usefull for
an easy check

I Asy K.F Th assuming that

- $V_{12} = 0$
- the system is A.S, so all Eig(F) are inside U.C

↓ Then

- ARE has 1! spd solution $\bar{P} \geq 0$
- DRE converges to $\bar{P} \quad \forall P_0 \geq 0$
- $\bar{K}$ corresponding, is such that all Eig $(F - \bar{K}H)$ are inside U.C

II Asy K.F Th assuming that

- $V_{12} = 0$
- (F, H) is fully OBSERVABLE (*)
- (F, Γ) is fully CONTROLLABLE (**)

↓ then

- ARE has 1! **pd** solution $\bar{P} > 0$
- DRE converges to $\bar{P} \quad \forall P_0 \geq 0$
- corresponding $\bar{K}$ is such that all Eig $(F - \bar{K}H)$ are inside U.C

(*) Fully OBSERVABILITY from
the OUTPUT y and only if

$$O = \begin{bmatrix} H \\ HF \\ \vdots \\ HF^{m-1} \end{bmatrix} \text{ is full Rank} \\ \text{Rank}(O) = m$$

(**) full CONTROLLABILITY from the noise $U_1(t)$

if $R = [\Gamma \quad F\Gamma \quad \dots \quad F^{m-1}\Gamma]$ is full rank

where here $U_1(t) \sim WN(0, V_1)$

we can write $V_1 = \Gamma \Gamma^T$ so that

$$S(t) = \Gamma U_1(t) \quad S(t) \sim WN(0, I)$$

Extended systems Sometimes one of the noises of the model system

can be different from a WN, for example

$$\begin{cases} x(t+1) = a x(t) + \eta(t) \\ y(t) = b x(t) + v_2(t) \end{cases} \quad \text{BUT } v_2 \sim WN(0,1)$$

$$\eta(t) = \frac{1}{1-cz^{-1}} e(t) \quad e(t) \sim WN(0,1) \quad \text{with } e \perp v_2$$

↓ we cannot apply K.F here! so we rewrite the system..

$$\eta(t) = c \eta(t-1) + e(t) \Rightarrow \eta(t+1) = c \eta(t) + \underbrace{e(t+1)}_{v_{12}(t) \sim WN(0,1)} \quad v_1 \perp v_2$$

we extend the states $x(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$ $x_1(t) = x(t)$ $x_2(t) = \eta(t)$

$$\begin{cases} x_1(t+1) = a x_1(t) + x_2(t) \\ x_2(t+1) = c x_2(t) + v_{12}(t) \\ y(t) = b x_1(t) + v_2(t) \end{cases} \quad \text{to this } F = \begin{bmatrix} a & 1 \\ 0 & c \end{bmatrix} \quad H = \begin{bmatrix} b & 0 \end{bmatrix}$$

we can apply K.F!

$$v_1(t) = \begin{pmatrix} 0 \\ v_{12}(t) \end{pmatrix} \sim WN(0, V_1) \quad V_1 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$v_2(t) \sim WN(0, v_2) \quad v_2 = 1 \quad v_{12} = 0$$

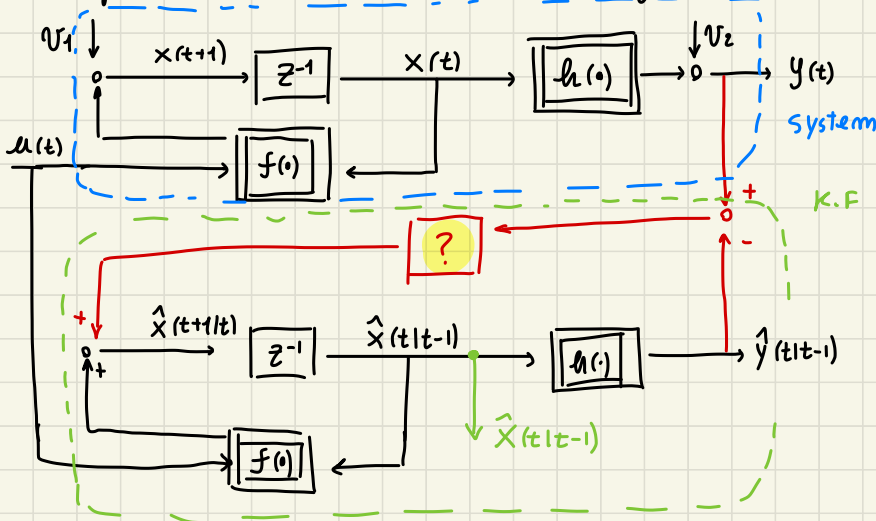
EXTENDED K.F extension for **NON LINEAR** systems

in a ss description

$$\begin{cases} x(t+1) = f(x(t), u(t)) + v_1(t) \\ y(t) = h(x(t)) + v_2(t) \end{cases}$$

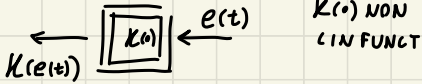
With $f(\cdot)$, $h(\cdot)$ **NON LINEAR** functions, of class C^1

We design K.F structure as classical one using "STATE OBSERVER"

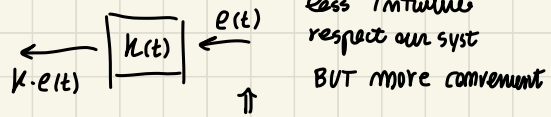


the issue is related to the **K. GAIN**? we can use two approaches

1 NON LIN. function Gain



2 LINEAR Time Varying GAIN Block



IDEA EKF make a local linear approx of the NON LIN syst $\forall$ time instance: NLT $\rightarrow$ LTV and use KF for LTV systems

$$\begin{cases} K(t) = (F(t)P(t)H^T(t) + V_2)^{-1} \cdot (H(t)P(t)H^T(t) + V_2)^{-1} \\ P(t+1) = (F(t)P(t)F^T(t) + V_1) - (F(t)P(t)H^T(t) + V_2)(H(t)P(t)H^T(t) + V_2)^{-1}(F(t)P(t)H(t) + V_1)^T \end{cases}$$

How to $\downarrow$

identify time varying **MATRIX**? Linear approx around $\hat{x}(t|t-1)$

$$F(t) = \left. \frac{\partial f(x(t), u(t))}{\partial x(t)} \right|_{\hat{x}(t|t-1)}$$

$$H(t) = \left. \frac{\partial h(x(t))}{\partial x(t)} \right|_{\hat{x}(t|t-1)}$$

$\downarrow$
so $F(t), H(t)$ are recomputed $\forall t$ (time instance) as local approx of f, h @ $\hat{x}(t|t-1)$

(EKF) ALG:

@ t:

- take last prediction $\hat{x}(t|t-1)$
- evaluate $F(t), H(t)$
- compute $K(t)$ and update $P(t) \rightarrow P(t+1)$
- compute new prediction $\hat{x}(t+1|t)$

← ITERATE

Comments

- EKF is a BIG extension of KF for NONLIN syst
- obviously EKF does NOT have a STEADY STATE solution (because it is an LTV syst)

• Issue

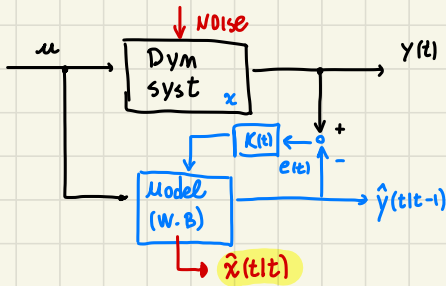
- stability impossible to guarantee
- computationally demanding

$\Downarrow$ very used BUT

- NOT in CRITICAL safety
- NOT in mission - CRITICAL
- high resources needed

Ch 2.5 SW SENSING with B.B approach

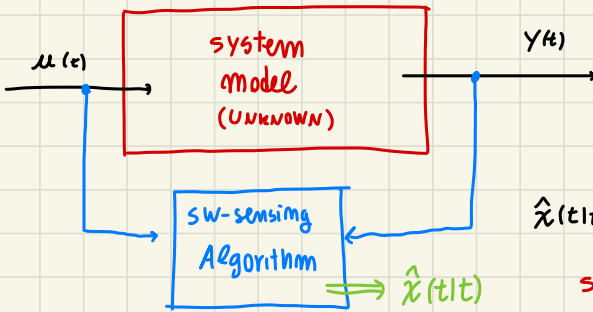
The SW sensing approach seen on K.F is the following



- needs a white box model of the physical system
- uses a feedback scheme, with a constructive Method
- In principle we don't need Training Dataset (in Reality maybe needed for V_1, V_2, V_3 estimation)
- usable for Totally un-measurable states

↓ can he used a different approach?
YES, BLACK BOX

LTI systems



Being an LTI system, our SW sensor is also LTI, and our objective is to identify the T.F of u, y to x dynamic

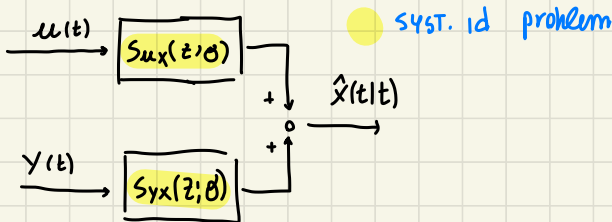
$$\hat{x}(t|t) = \underbrace{S_{ux}(z)}_{\text{system identification problem}} u(t-1) + \underbrace{S_{yx}(z)}_{\text{system identification problem}} y(t)$$

STEP #1) Dataset collection

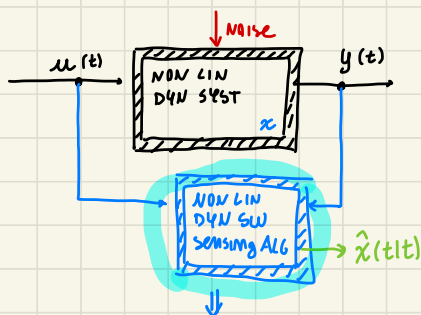
$$\{u(1), \dots, u(N)\} \quad \{y(1), \dots, y(N)\} \quad \{x(1), \dots, x(N)\}$$

↑
The states to estimate need to be measurable during the training phase, then on production phase (after train) are estimable

STEP #2 ... etc) This is simply a syst id. problem that can be approached by different seen algorithm like LSID OR a model based approach



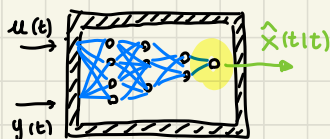
NON LINEAR SYSTEMS



here you still have a syst id. problem from the dataset as before

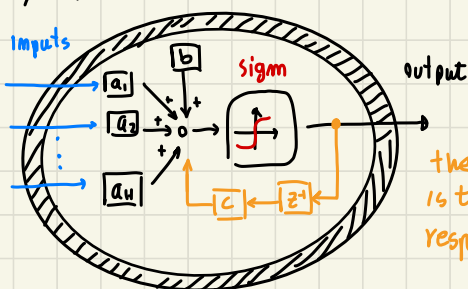
With multiple possible architecture solutions: of the SW sensing Alg.

ARCH 1 (most classical) dynamical Neural network



after training phase from Dataset you get the parameters a_i, b_i, c_i of all the neurons of the net

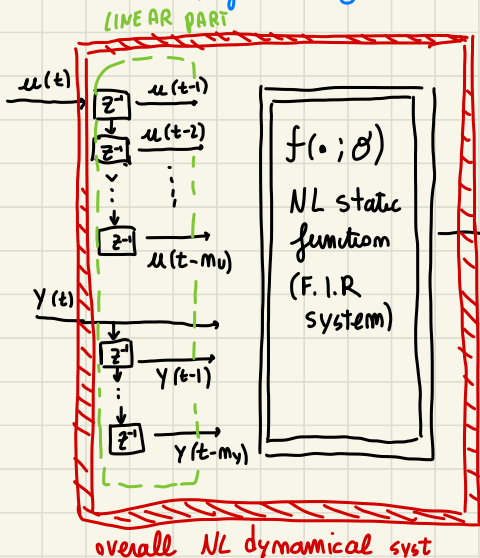
focus on each neuron of a dym NN



the recursive part is the difference respect a STATIC N.N

ISSUE:
 • computationally demanding Training
 • No guarantee of convergence (stability)
 } for this, that general each. is NOT the most used!

ARCH 2 simplification of (1), use a combination of LINEAR + NL static FIR syst



With this "SPLITTING" of the NL dym system:

• Training is done on a much simpler system (NL static N.N $\rightarrow$ standard train)
 • here $f(\cdot; \theta)$ a F.I.R filter, it is as. stable for some $\rightarrow$ guarantee of stability

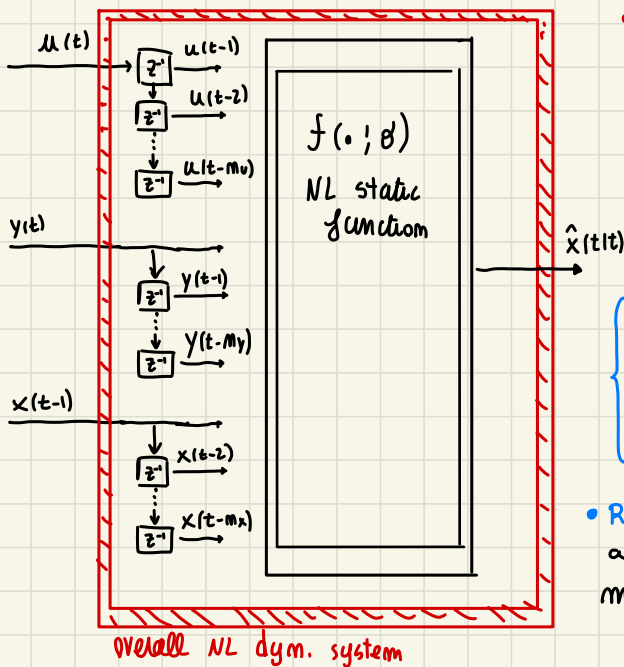
ISSUE high dimension of $f(\cdot; \theta)$

$$f: \mathbb{R}^{(m \cdot m_u + p \cdot m_y)} \rightarrow \mathbb{R}^m$$

can be very high dim. for a MIMO system!

$\rightarrow$ to solve this...

ARCH 3 Qs (2) But use RECURSIVENESS, so an IIR scheme

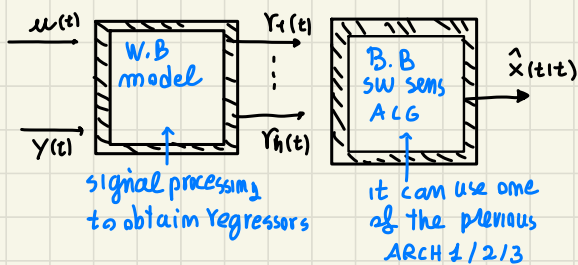


- the issue is the RECURSIVENESS, so NO guarantee of stability
- but Respect ARCH 2 here having also the recursive part we need less input, so f is even less respect previously

{ during training we feedback the x measurement, in practice instead $\hat{x}(t-1|t-1), \dots$ previous estimation }

- Respect ARCH1 it need a training of a STATIC NN, so much simpler

ARCH 4 (modification of previous one) SPLIT the problem in 2 steps



- the amount "h" of regressors is much smaller than the overall inputs "p + m"

- If correctly tuning 1st stage, the 2nd one could be very simple

ISSUE: the tuning of REGRESSORS is NOT simple! there is NOT a general rule

SW SENSING COMPARISON WB(KF) VS BB

PRO CONS

comparison features	Needling of a DATASET	possibility to measure UN-measurable variables	easy re-tuning for new syst.	quality of estimation	Interpretation of the scheme	Need of a WB model of syst
Model based K.F	NO, constr. method (in reality YES for V_1, V_2)	YES, no training Data of it needed	YES, Just recompute the gain	GOOD but limited by model	EASY, just digital twin and correction	YES, based all on the availability of that model
Black Box	YES, ML procedure for Training	NO, you need measure to TRAIN.	NO, you need to re-train all the syst	VERY GOOD, does NOT depend on syst.	HARD, NN acting with complex str.	NO, just Black Box procedure

ch 2.6 (Hybrid) GRAY-BOX system identification

(usage of K.F. with sensing for model identification)

In an hybrid framework where we have the knowledge about the system as model description BUT we miss some information

$$\begin{cases} x(t+1) = f(x(t), u(t), \theta) + v_1(t) \\ y(t) = h(x(t), \theta) + v_2(t) \end{cases}$$

↓ we use the **EXTENDED** system trick of K.F

$f(\cdot)$, $h(\cdot)$ are known function found by a physical analysis of the system, where most of model parameters are known, while some θ unknown must be estimated from data

$$\begin{cases} x(t+1) = f(x(t), u(t), \theta(t)) + v_1(t) \\ \theta(t+1) = \theta(t) + v_\theta(t) \\ y(t) = h(x(t), \theta(t)) + v_2(t) \end{cases}$$

$$x_\varepsilon(t) = \begin{pmatrix} x(t) \\ \theta(t) \end{pmatrix} \quad v_{\varepsilon_1}(t) = \begin{pmatrix} v_1(t) \\ v_\theta(t) \end{pmatrix} \quad v_2(t)$$

even if usually we deal with constant parameters $\bar{\theta}$, we need to introduce a noise even in the parameter so the K.F has a variation respect the initial value $\theta(0)$ and we converges to the correct values

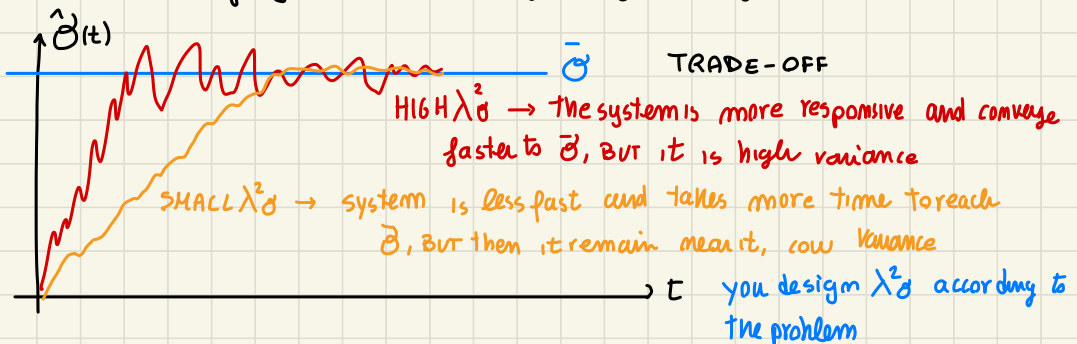
(even if this is unstable, no issue, K.F works also for unstable systems)

The issue is related to Definition of $v_\theta(t)$:

$v_\theta(t) \sim WN(0, V_\theta)$ with $v_\theta \perp v_1$, $v_\theta \perp v_2$ (UNCORRELATED)

and $V_\theta = \begin{bmatrix} \lambda_\theta^2 & & 0 \\ & \lambda_\theta^2 & \\ 0 & & \ddots \\ & & & \lambda_\theta^2 \end{bmatrix}$ so we just need to TUNE λ_θ^2 .
This can be done by DATA DRIVEN PROCEDURES (empirically by "TUNING KNOB")

We should be carefully when empirically tuning λ_θ^2 , in fact:



- NOTICE that this technique can be used for time-varying parameters $\theta(t)$, TRACKING run-time when system is Time varying

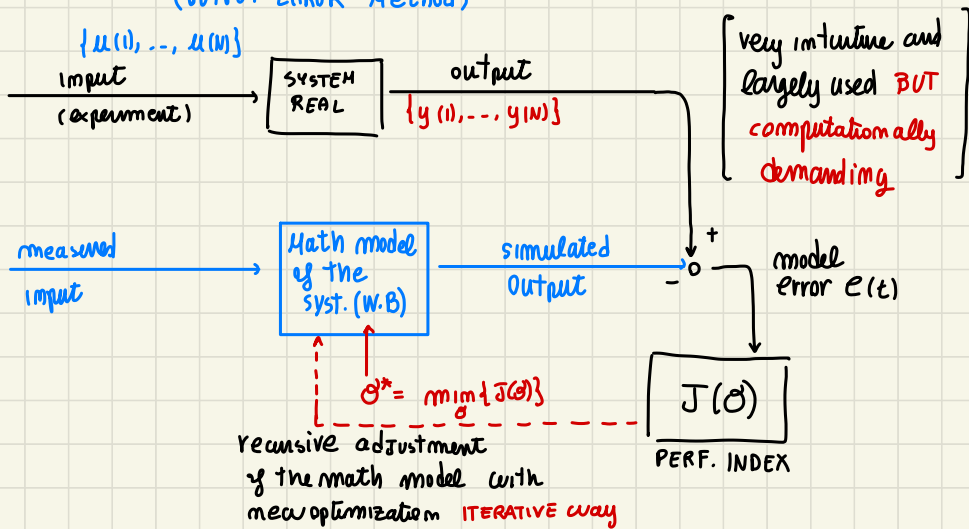
- **CAREFULL!** When using this procedure, IF the amount of STATES and PARAMETER is much larger than outputs it will work BADLY!

we need a limited $\frac{|\text{OUTPUT}|}{|\text{STATES}| + |\text{PARAM.}|}$ ratio

- **ISSUE** usually when using gray box id. the parameters $\theta(t)$ are system param. typically multiplexed to states, so you obtain a N.L dym system
 ↳ E.K.F for identification!

OTHER TECHNIQUE alternatives to GRAY BOX for parameters estimation

there are that **SIMULATION ERROR METHOD** to estimate system parameters (OUTPUT ERROR Method)



TRADE-OFF SEM vs GRAY-BOX (K.F)

IT is better to use **K.F** when you have MORE parameter θ to estimate OR when θ is TIME VARIING (Run-time update)

IT is better to use **S.E.M** for small amount of constant θ (OFF-line estim.)